[Assignment-1]

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Ans to the question no 1

a

1 Frame = (3840 × 2160) pixels
= 8294400 pixels

while,
FPS = 60,

pixels per second = 8294400 × 60
= 497 664 000 pixels/s

bits per second = (497 664 000 × 24)
= 1.194×10^{10} bits/s

b

A video runs at 3840×2160 resolution.

∴ 1 Frame = 8294400 pixels

The GPU processes 95 000 pixels /ms

∴ For the given video,

To process a frame, GPU takes $= \dfrac{8294400}{95000}$

= 87.31 ms

which is greater than 16.67 ms.

so, the GPU cannot render one full frame (3840× 2160) within 16.67 ms.

$\underline{\underline{c}}$

we know that,

$$\text{Frame size} = \text{Width} \times \text{Height} \times \text{Depth}$$

$$= \underbrace{\cancel{\text{Resolution}}}_{\text{Total pixels}} \times \text{Depth}$$

$$\Rightarrow 24883200 = \text{Total pixels} \times 24$$

$$\Rightarrow \text{Total pixels} = \frac{24883200}{24}$$

$$= (1036800 \times 8)$$

$$= 8294400$$

$\therefore$ pixels per frame = 8294400

and,

pixels per frame for $(3840 \times 2160) = 8294400$

$\therefore$ Both are same.

Ans to the question no 2

a

$$\text{Slope}, m = \frac{-5-(-12)}{0-5} = \frac{-5+12}{-5} = -\frac{7}{5}$$

(5,-12) to (0,-5)

Since, m is not in a range of -1 to 1.

$$y_{k+1} = y_k + 1$$

$$x_{k+1} = x_k \pm \frac{1}{m}$$

$$\frac{1}{m} = -\frac{5}{7} = -0.7$$

y(+1)	x(+ $\frac{1}{m}$)	x(round-off)	pixel
-12	5	5	(5,-12)
-11	4.3	4	(4,-11)
-10	3.6	4	(4,-10)
-9	2.9	3	(3,-9)
-8	2.1	2	(2,-8)
-7	1.4	1	(1,-7)
-6	0.7	1	(1,-6)
-5	0	0	(0,-5)

b

Given,

$7x - 4y + 56 = 0$

$\Rightarrow 7 \cdot 0 - 4y + 56 = 0$

$\Rightarrow 4y = 56$

$\therefore y = 14$

intersects the y axis.

$\therefore x = 0$

Let,

$\therefore$ Starting point $P_1 \equiv (0, 14)$

Ending point $P_2 \equiv (10, 5)$

$dx = 10 - 0 = 10$

$dy = 5 - 14 = -9$

$|dx| = 10$

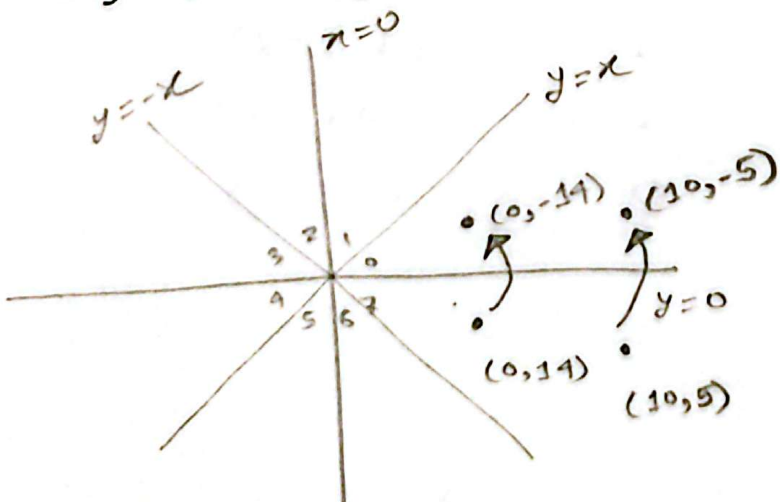
$|dy| = 9$

$\therefore dx > dy$

$dx > 0$ and $dy < 0$ $\}$ Q_4 ; since $|dx| > |dy|$ so, the zone is more closer to x axis.

$\therefore$ zone $= 7$

So, 8 way symmetry needed.



$\therefore P_1' \equiv (0, -14)$

$P_2' \equiv (10, -5)$

$dy' = 9$

$dx' = 10$

$$d_{init} = 2dy' - dx' = 2 \times 9 - 10 = 8$$

$$\Delta NE = 2dy' - 2dx' = 2 \times 9 - 2 \times 10 = -2$$

$$\Delta E = 2dy = 2 \times 9 = 18$$

x'	y'	D	Directions (NE/E)	x	y	pixel
0	-14	8	NE	0	14	(0,14)
1	-13	6	NE	1	13	(1,13)
2	-12	4	NE	2	12	(2,12)
3	-11	2	NE	3	11	(3,11)
4	-10	0	E	4	10	(4,10)
5	-10	18	NE	5	10	(5,10)
6	-9	16	NE	6	9	(6,9)